

The MQ-Calculus:

A Calculus for Quantum Processes Involving Intermittent Measurement

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1 Introduction and Motivation

We introduce a calculus for processes that involve measurements interspersed throughout the computation. The key intuition is that *communication* = *measurement*. In the calculus presented below this is represented explicitly in the **COMM** rule.

2 Related Work

The calculus is related to the CQP calculus of Gay and Nagarajan [1]. The primary difference is that their calculus has an explicit measurement operator, whereas this calculus insists that measurement happens when a quantum state is communicated from one agent to another.

3 Technical Presentation

3.1 Syntax

$$\theta ::= [0, 2\pi)$$

States.

$$\begin{aligned} S, T ::= & |0\rangle \mid |1\rangle \mid v \\ & \mid H(S) \mid F(\theta)(S) \mid R(\theta)(S) \\ & \mid \text{CTRL-}F(\theta)(S, T) \mid \text{CNOT}(S, T) \mid \text{CTRL}(U)(S, S, S, \dots)(T, T, T, \dots) \\ & \mid S \otimes T \mid \mathbb{C} S \mid S + T \end{aligned}$$

Processes.

$$Q ::= \bigotimes_j x_j!(S) \quad (\text{output; when } j \text{ ranges over an empty domain this is } \mathbf{0})$$

$$P ::= \text{for} \left(\bigotimes_j m_j \leftarrow \bigotimes_j x_j \right) \{Proc\} \quad (\text{input; when } j \text{ ranges over an empty domain this is } \mathbf{0})$$

$$Proc ::= Q \mid P \mid \sum_i p_i Proc \mid Proc \mid Proc \quad \text{where } p_i \in \mathbb{R}^+$$

3.2 Hypothesis

$$p(P_1 \mid P_2) = p P_1 \mid p P_2$$

3.3 Distributing Sums over Parallel Composition

Note that we have

$$\begin{aligned} & \sum_{i_1} p_{1,i_1} \left(\bigotimes_j x_{i_1,j}!(S_{i_1}) \right) \mid \sum_{i_2} p_{2,i_2} \left(\mathbf{for} \left(\bigotimes_j m_{i_2,j} \leftarrow \bigotimes_j x_{i_2,j} \right) \{Proc_{i_2}\} \right) \\ &= \sum_{i_1, i_2} p_{1,i_1} p_{2,i_2} \left(\bigotimes_j x_{i_1,j}!(S_{i_1}) \right) \mid \left(\mathbf{for} \left(\bigotimes_j m_{i_2,j} \leftarrow \bigotimes_j x_{i_2,j} \right) \{Proc_{i_2}\} \right) \end{aligned}$$

This will be useful in writing the full comm rule.

3.4 The Comm Rule

We build the comm rule in steps. The driving intuition is that

$$p_1 \bigotimes_{j \in J} x_j!(S) \mid p_2 \mathbf{for} \left(\bigotimes_{j \in J} m_j \leftarrow \bigotimes_{j \in J} x_j \right) \{Proc\} \longrightarrow p_1 p_2 \sum_r |\langle r | S \rangle|^2 Proc\{r/m\}$$

where S is a $|J|$ -qubit state.

Thus the comm rule in full is:

Comm:

$$\frac{\sum_{i_1} p_{1,i_1} \left(\bigotimes_{j_1} x_{i_1,j_1}!(S_{i_1}) \right) \mid \sum_{i_2} p_{2,i_2} \left(\mathbf{for} \left(\bigotimes_{j_2} m_{i_2,j_2} \leftarrow \bigotimes_{j_2} x_{i_2,j_2} \right) \{Proc_{i_2}\} \right)}{\sum_{i_1, i_2} p_{1,i_1} p_{2,i_2} \sum_r |\langle r | S_{i_1} \rangle|^2 Proc_{i_2}\{r/m_{i_2}\}} \bigotimes_j x_{i_1,j} = \bigotimes_j x_{i_2,j}$$

where the outer sum ranges over all pairs (i_1, i_2) such that the channel tuples match, i.e. $\bigotimes_j x_{i_1,j} = \bigotimes_j x_{i_2,j}$, and the inner sum over r ranges over the $|J|$ -qubit computational basis states. For each such pair, the probability weight is the product $p_{1,i_1} p_{2,i_2}$, the state S_{i_1} is measured in the computational basis yielding outcome r with probability $|\langle r | S_{i_1} \rangle|^2$, and the continuation $Proc_{i_2}\{r/m_{i_2}\}$ is the body of the input with the measurement outcome substituted for the bound pattern variables.

4 Examples

1. Simple communication:

$$x!(S) \mid \mathbf{for}(s \leftarrow x)\{P\}$$

2. Iterated Hadamard:

$$x!(|0\rangle) \mid \mathbf{for}(s \leftarrow x) \left\{ x!(Hs) \mid \mathbf{for}(s \leftarrow x) \{x!(Hs)\} \right\}$$

3. Collapsed form:

$$x!(|0\rangle) \mid \mathbf{for}(s \leftarrow x)\{x!(H H s)\}$$

4. General circuit form:

$$wires!(|0 \cdots 0\rangle) \mid \mathbf{for}(initialState \leftarrow wires)\{output!(U_{\text{circuit}} initialState)\}$$

4.1 Notational Conventions

$$\begin{aligned} 1\ x!(0) \otimes 1\ y!(0) &= 1\ (xy)!(00) \\ (H \otimes I)(1\ x!(0) \otimes 1\ y!(0)) &= \frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(10) \\ \text{CNOT}\left(\frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(10)\right) &= \frac{1}{\sqrt{2}}\ (xy)!(00) + \frac{1}{\sqrt{2}}\ (xy)!(11) \\ \text{CNOT}\left((H \otimes I)(1\ x!(0) \otimes 1\ y!(0))\right) &\mid \mathbf{for}(v \leftarrow x)\{X\} \mid \mathbf{for}(w \leftarrow y)\{Y\} \end{aligned}$$

5 Analysis of Wigner's Friend Paradox

We model the Wigner's friend thought experiment in the MQ-calculus. Three agents—the cat, the friend, and Wigner—each hold one qubit.

Setting up the cat state.

$$cat!(H |0\rangle) \quad // \text{ sets up cat state}$$

Note that in the literature $H |0\rangle |0\rangle |0\rangle$ is also written $|+00\rangle$. Combining the three single-qubit outputs:

$$cat!(H |0\rangle) \otimes friend!(|0\rangle) \otimes wigner!(|0\rangle) = (cat \otimes friend \otimes wigner)!(H |0\rangle |0\rangle |0\rangle)$$

MWI (no collapse). Under a many-worlds interpretation, no measurement occurs and the unitary propagates through all three qubits:

$$(cat \otimes friend \otimes wigner)!(\text{CNOT}_{friend, wigner} \text{CNOT}_{cat, friend} H |0\rangle |0\rangle |0\rangle)$$

Copenhagen (sequential measurement). Under a Copenhagen interpretation, each communication step triggers a measurement. In the MQ-calculus this is expressed as nested input/output:

$$\begin{array}{l}
(cat \otimes friend \otimes wigner)!(H |0\rangle |0\rangle |0\rangle) \\
| \\
\mathbf{for}(catState, friendState \leftarrow cat \otimes friend) \{ \\
\quad friend!(\text{CNOT}_{cat, friend} |catState\rangle |friendState\rangle) \\
\quad | \\
\quad \mathbf{for}(friendState, wignerState \leftarrow friend \otimes wigner) \{ \\
\quad \quad wigner!(\text{CNOT}_{friend, wigner} |friendState\rangle |wignerState\rangle) \\
\quad \} \\
\}
\end{array}$$

The key distinction is that in the MWI version, the entire circuit is applied unitarily to the joint state before any output, while in the Copenhagen version, each **for**($\dots$) constitutes a measurement: the comm rule fires, the state collapses, and subsequent operations act on the post-measurement state. The MQ-calculus makes this difference syntactically explicit. That is, in the MQ-calculus there is no paradox. There are two different, but equally valid programs, and either may be executed.

6 Alternative Design Options

6.1 Classical Processes Acting on Quantum States

A configuration consists of:

1. A number n of allocated wires.
2. A 2^n -dimensional state $|\psi\rangle$.

A *quantum process* acts on a configuration.

6.2 De Bruijn Indices

$$\begin{array}{l}
P, Q ::= \mathbf{0} \\
\quad | P \mid Q \\
\quad | \mathbf{new} P \\
\quad | U[i_0, \dots, i_{k-1}] ; P \\
\quad | \mathbf{out} i \\
\quad | \mathbf{in} i \{P, Q\}
\end{array}$$

Shift operation.

$$\begin{aligned}
\text{shift}(c, \mathbf{new} P) &= \mathbf{new} \text{shift}(c + 1, P) \\
\text{shift}(c, P \mid Q) &= \text{shift}(c, P) \mid \text{shift}(c, Q) \\
\text{shift}(c, U[is]; P) &= U[\text{shift}(c, is)]; \text{shift}(c, P) \\
\text{shift}(c, \mathbf{out} i) &= \mathbf{out} \text{shift}(c, i) \\
\text{shift}(c, \mathbf{in} i \{P, Q\}) &= \mathbf{in} \text{shift}(c, i) \{\text{shift}(c, P), \text{shift}(c, Q)\}
\end{aligned}$$

6.3 Denotational Semantics

These processes act linearly on the distribution:

$$\llbracket U[i_0, \dots, i_{k-1}]; P \rrbracket(n, |\psi\rangle) = \llbracket P \rrbracket(n, U|\psi\rangle)$$

where $U|\psi\rangle$ is shorthand for acting on those qubits.

$$\llbracket \mathbf{new} P \rrbracket(n, |\psi\rangle) = \llbracket P \rrbracket(n + 1, |0\rangle \otimes |\psi\rangle)$$

Parallel composition (interleaving semantics).

$$\llbracket \prod_i P_i \rrbracket(n, |\psi\rangle) = \text{random permutation } \sigma \left(\bigcirc_{\sigma(i)} \llbracket P_{\sigma(i)} \rrbracket \right)(n, |\psi\rangle)$$

$$\llbracket P \rrbracket(n, |\psi\rangle) \otimes \llbracket Q \rrbracket(m, |\phi\rangle) \longrightarrow \llbracket P \mid \text{shift}(Q)^n \rrbracket(n + m, |\psi\rangle |\phi\rangle)$$

Synchronization is measurement.

$$\llbracket \mathbf{out} i \mid \mathbf{in} i \{P, Q\} \rrbracket(n, |\psi\rangle) = \Pr(i=|0\rangle) \llbracket P \rrbracket(n, |\psi\rangle_{0/i}) + \Pr(i=|1\rangle) \llbracket Q \rrbracket(n, |\psi\rangle_{1/i})$$

where $\{x/i\}$ means we collapse qubit i of ψ to $|x\rangle$.

6.4 Structural Congruence

The usual equations hold:

$$\begin{aligned}
P \mid \mathbf{0} &= P \\
P \mid Q &= Q \mid P \\
(P \mid Q) \mid R &= P \mid (Q \mid R) \\
\mathbf{new} P \mid Q &= \mathbf{new} (P \mid \text{shift}(1, Q))
\end{aligned}$$

7 Conclusions and Future Work

References

- [1] Simon J. Gay and Rajagopal Nagarajan. Communicating quantum processes. In Jens Palsberg and Martín Abadi, editors, *Proceedings of the 32nd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages, POPL 2005, Long Beach, California, USA, January 12-14, 2005*, pages 145–157. ACM, 2005.